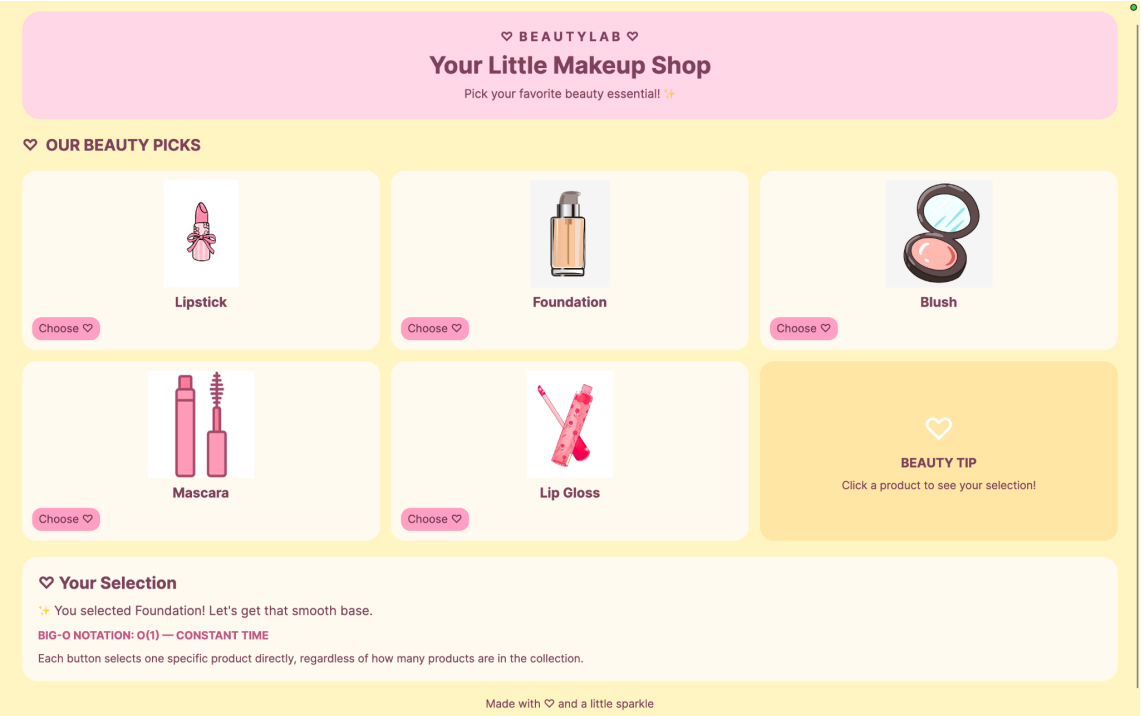
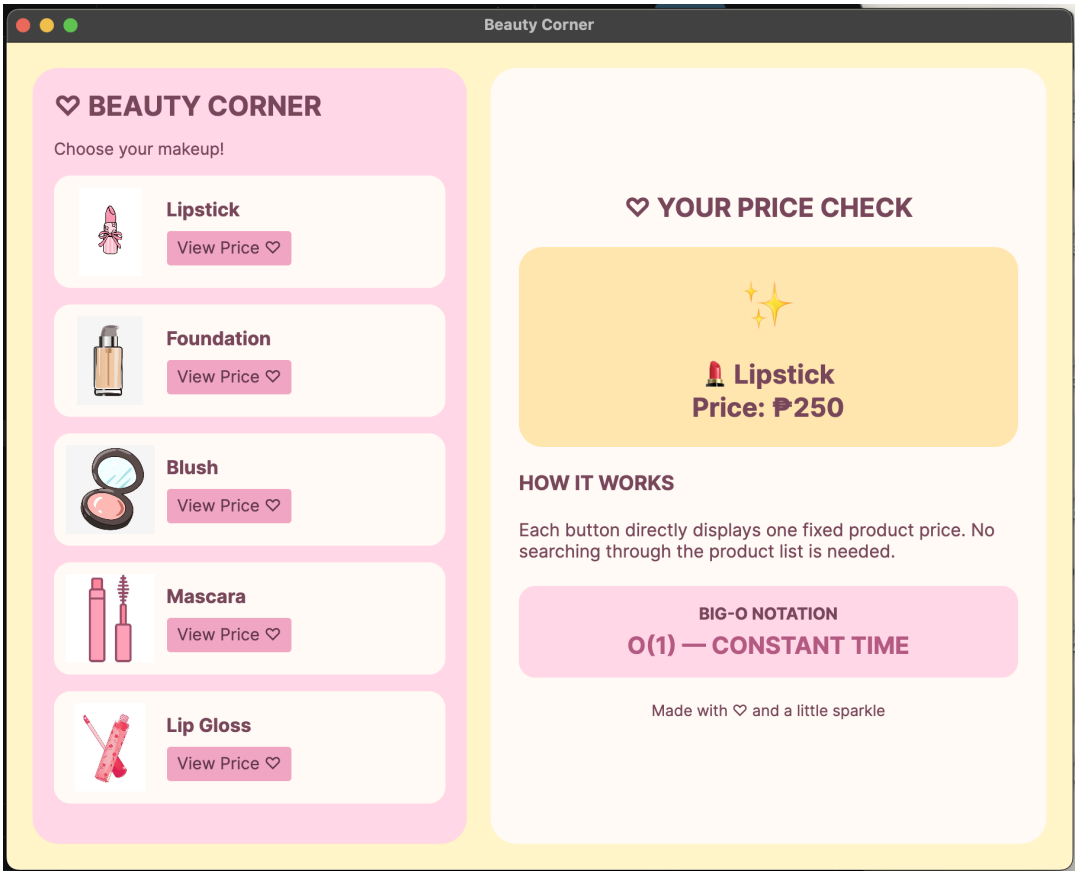
Constant Time





Beauty Sale ♡ Discount Calculator

♡BEAUTYLAB♡

Beauty Sale Calculator

Save more on your makeup favorites! ✨

♡ SELECT PRODUCT



Everyday Lipstick

Original price: ₱250

Fixed discount: 10%

Apply 10% Discount ♡

♡ YOUR SAVINGS

Discount Amount

₱25.00

Final Price

₱225.00

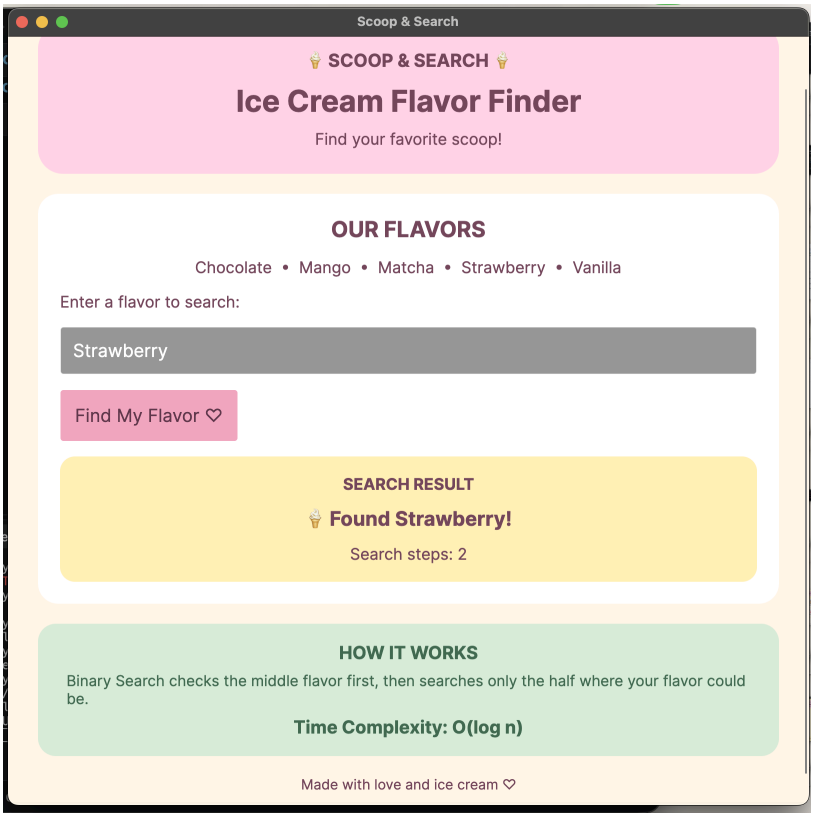
Discount applied successfully! You saved ₱25.00. ♡

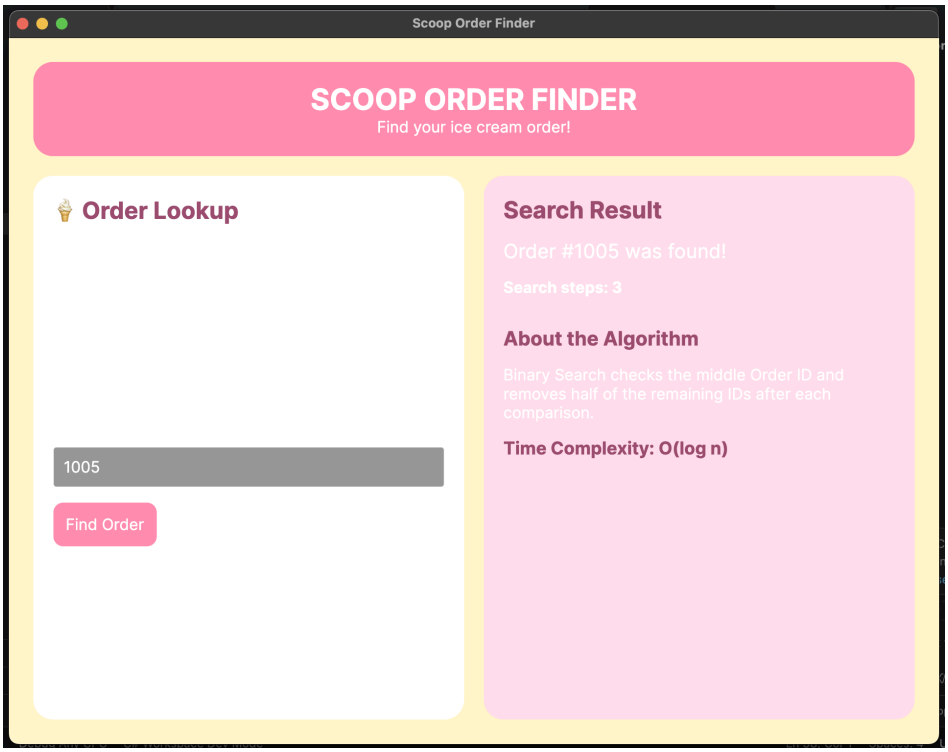
BIG-O NOTATION: O(1)

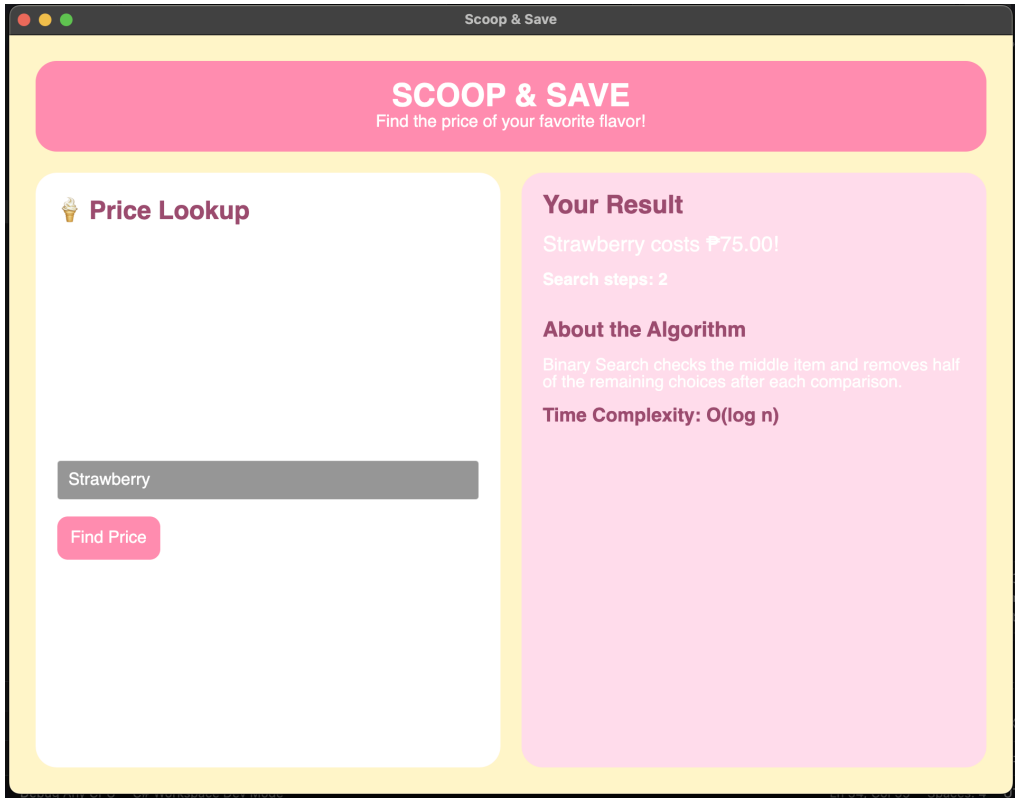
The program calculates a fixed discount using a constant number of arithmetic operations. The amount of work does not depend on the number of products.

Made with ♡ and a little sparkle

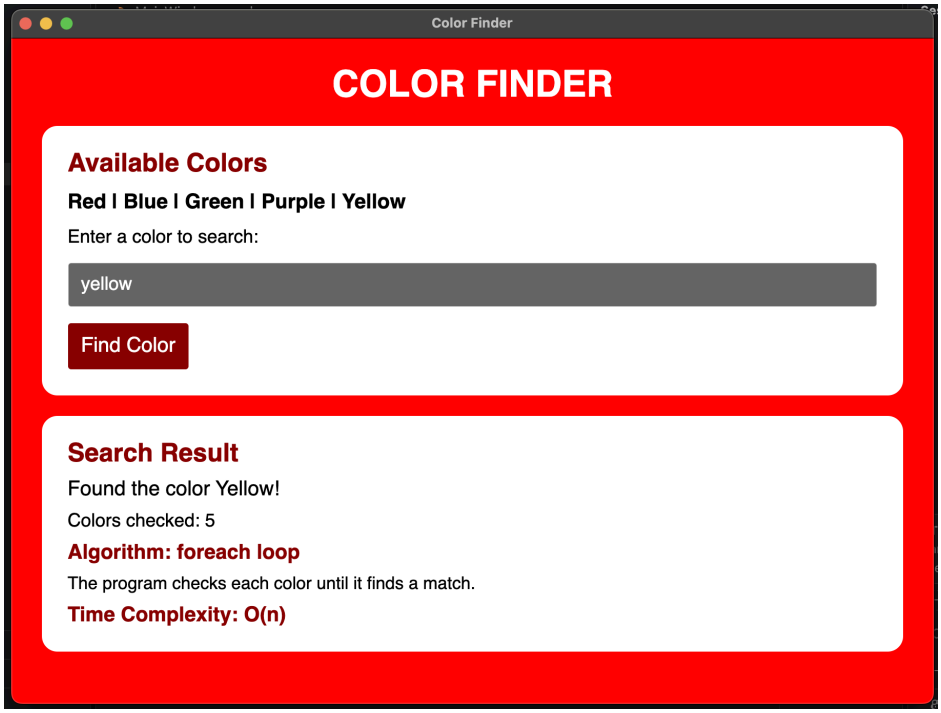
Logarithmic Time

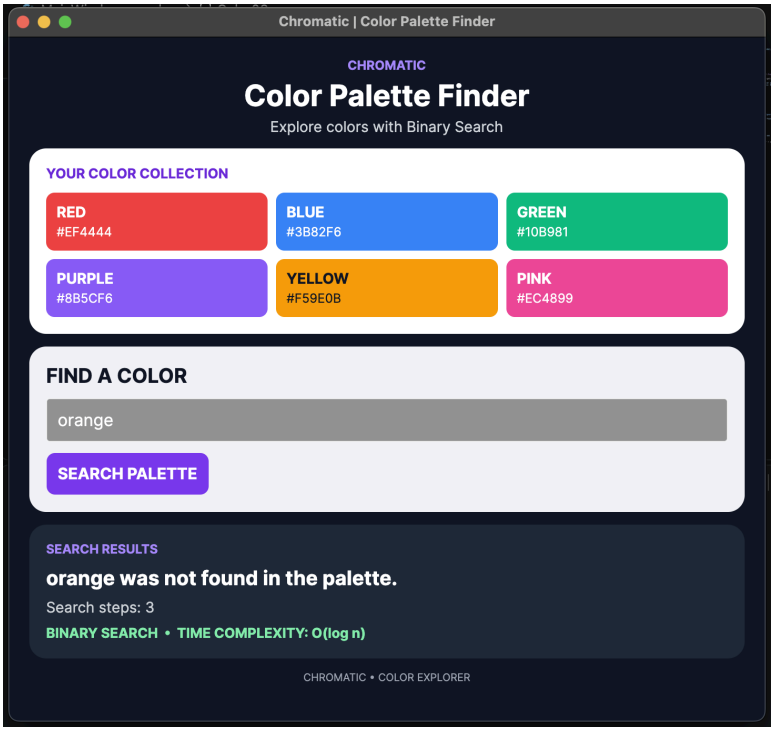


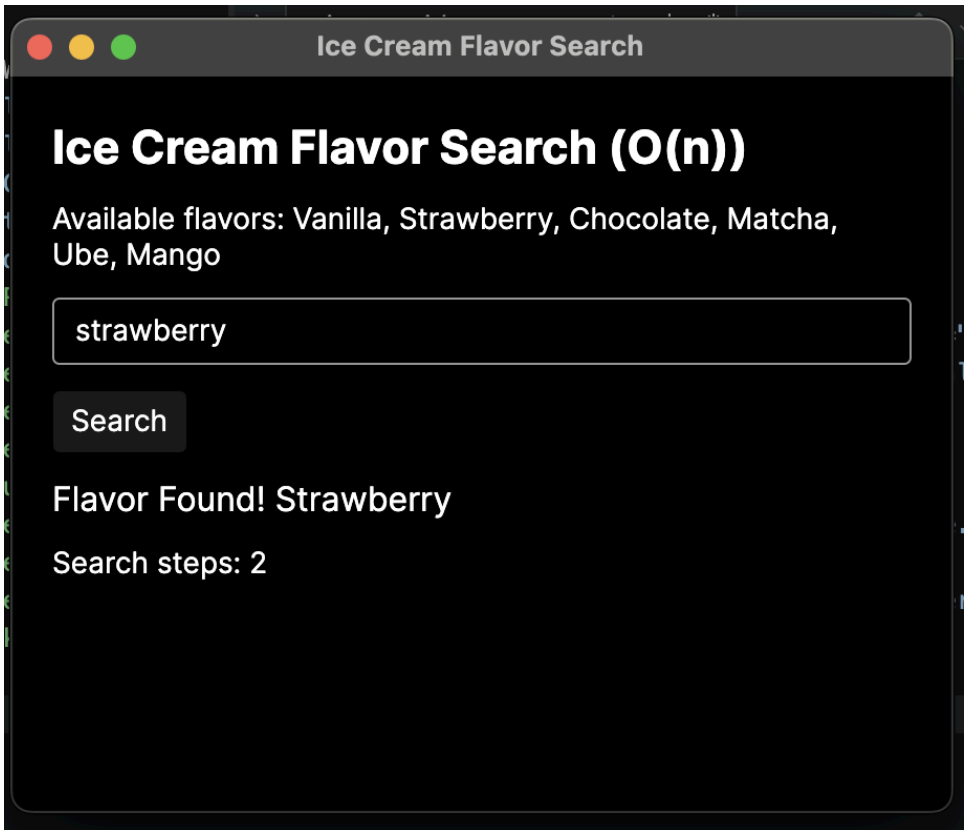



Scoop & Save
SCOOP & SAVE
Find the price of your favorite flavor!
Price Lookup
Strawberry
Find Price
Your Result
Strawberry costs ₱75.00!
Search steps: 2
About the Algorithm
Binary Search checks the middle item and removes half of the remaining choices after each comparison.
Time Complexity: O(log n)

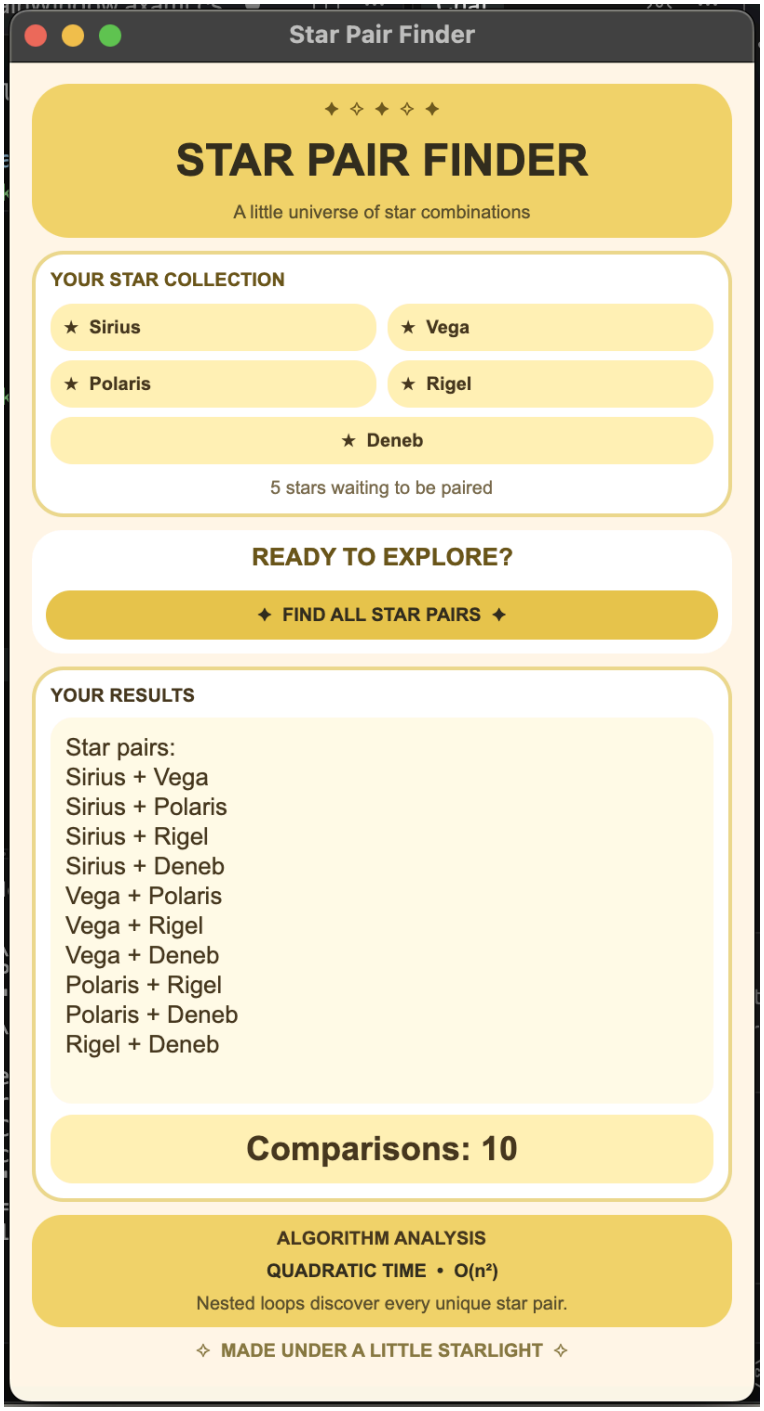
Linear

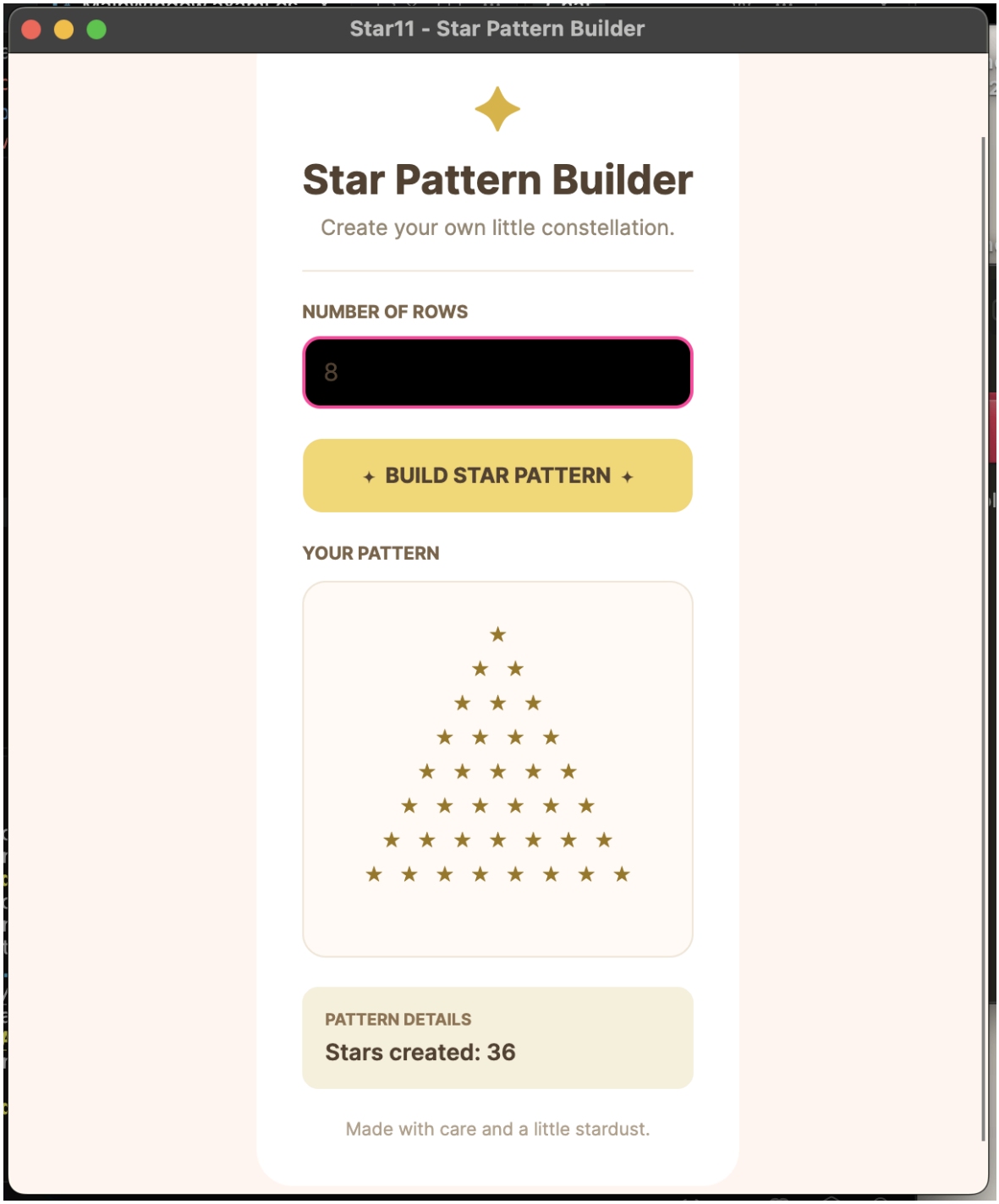

Color Finder
COLOR FINDER
Available Colors
Red | Blue | Green | Purple | Yellow
Enter a color to search:
yellow
Find Color
Search Result
Found the color Yellow!
Colors checked: 5
Algorithm: foreach loop
The program checks each color until it finds a match.
Time Complexity: O(n)

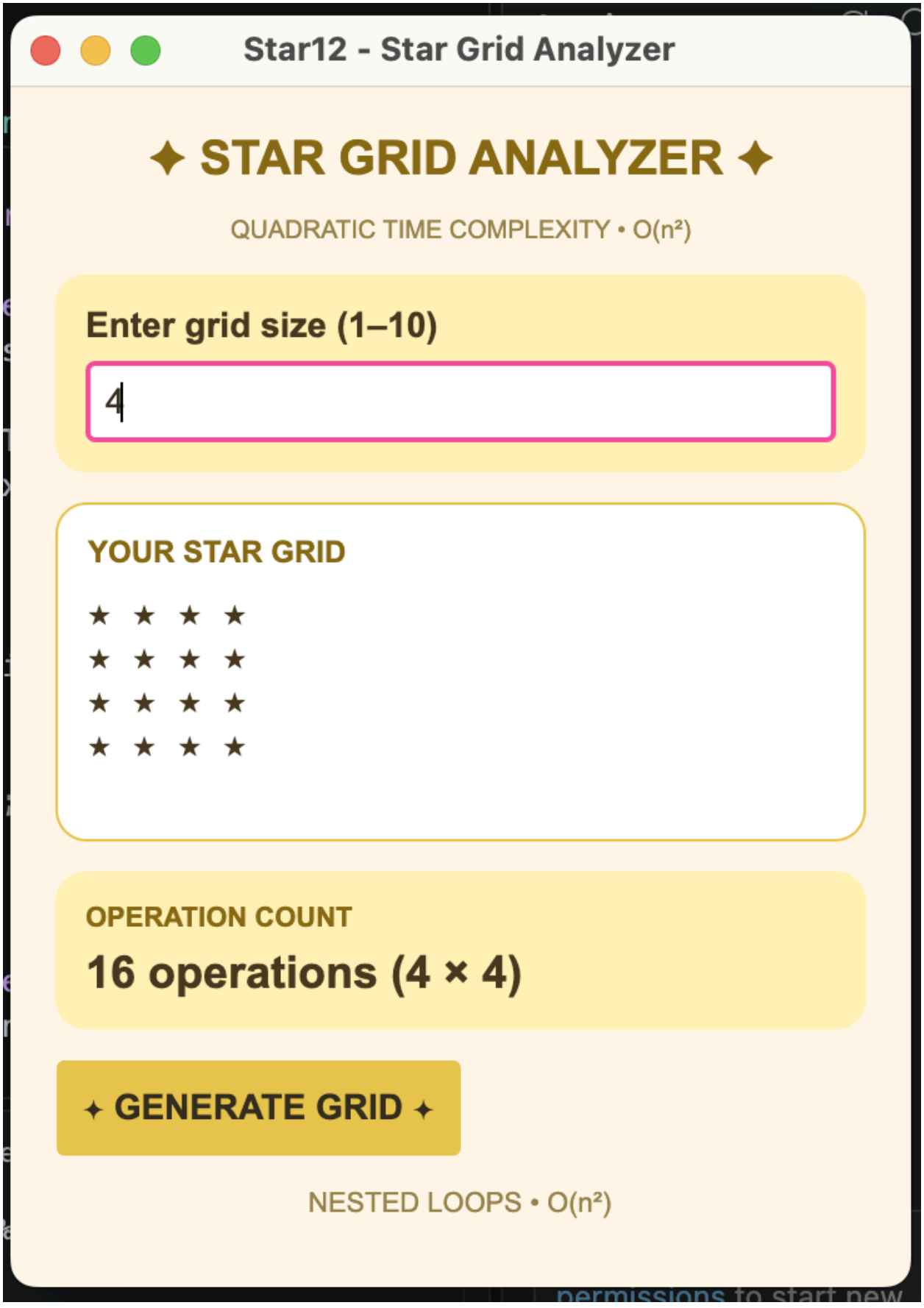

Chromatic | Color Palette Finder
CHROMATIC
Color Palette Finder
Explore colors with Binary Search
YOUR COLOR COLLECTION
RED
#EF4444
BLUE
#3B82F6
GREEN
#10B981
PURPLE
#8B5CF6
YELLOW
#F59E0B
PINK
#EC4899
FIND A COLOR
orange
SEARCH PALETTE
SEARCH RESULTS
orange was not found in the palette.
Search steps: 3
BINARY SEARCH • TIME COMPLEXITY: O(log n)
CHROMATIC • COLOR EXPLORER


Ice Cream Flavor Search
Ice Cream Flavor Search (O(n))
Available flavors: Vanilla, Strawberry, Chocolate, Matcha, Ube, Mango
strawberry
Search
Flavor Found! Strawberry
Search steps: 2

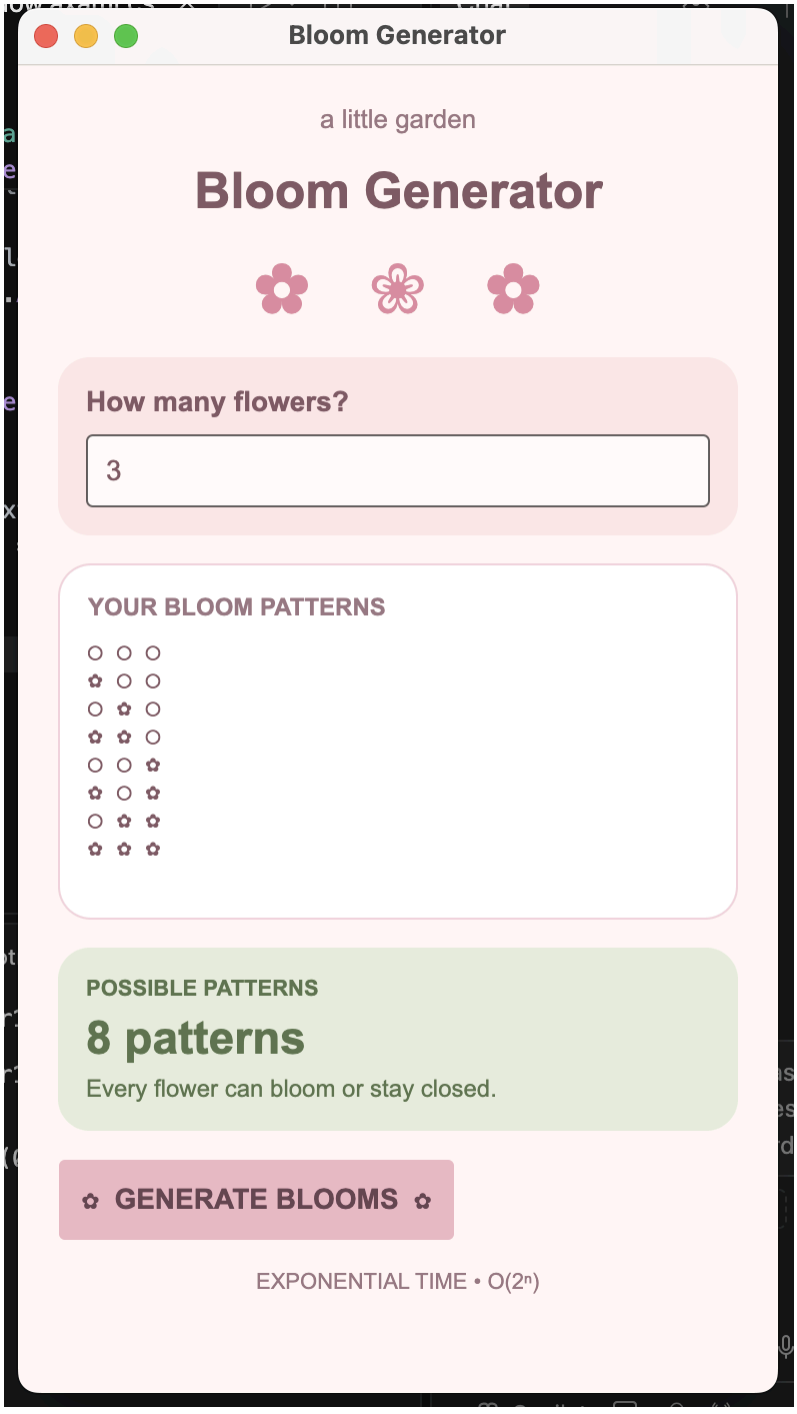
Quadratic Time

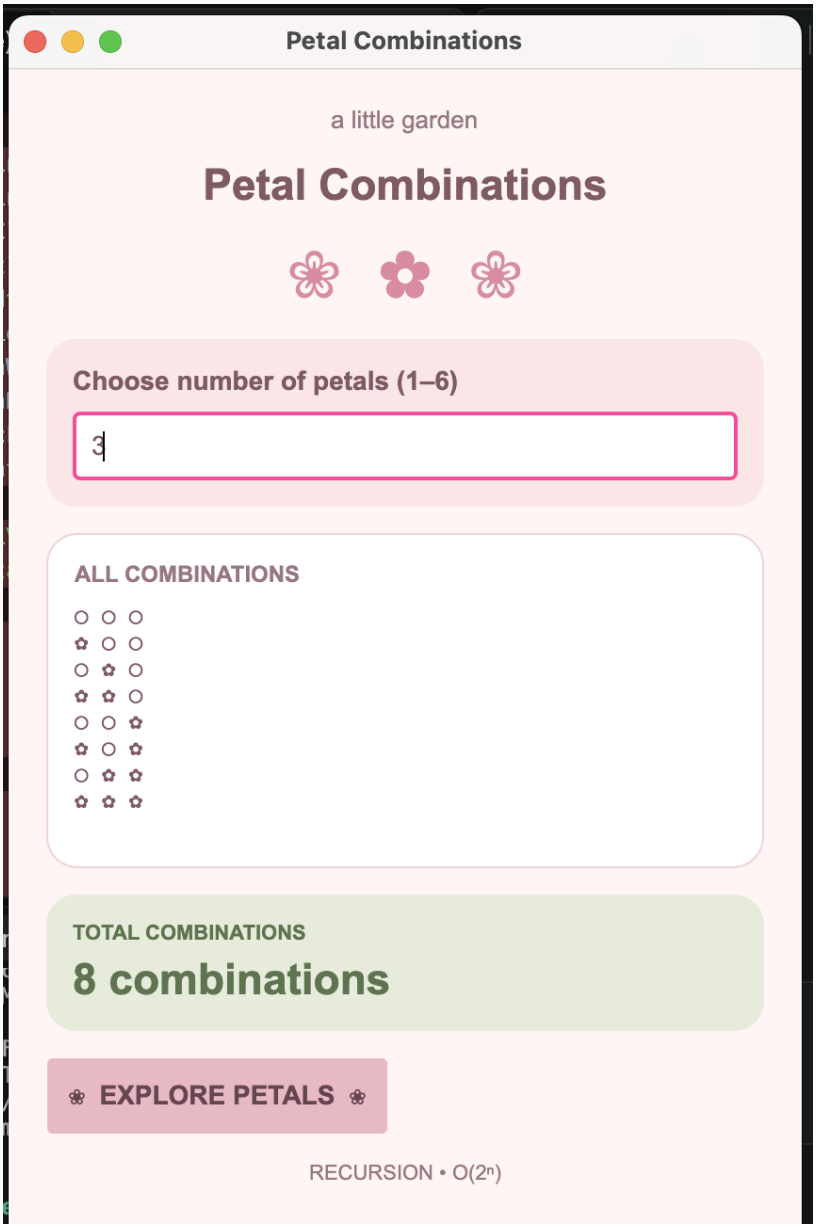


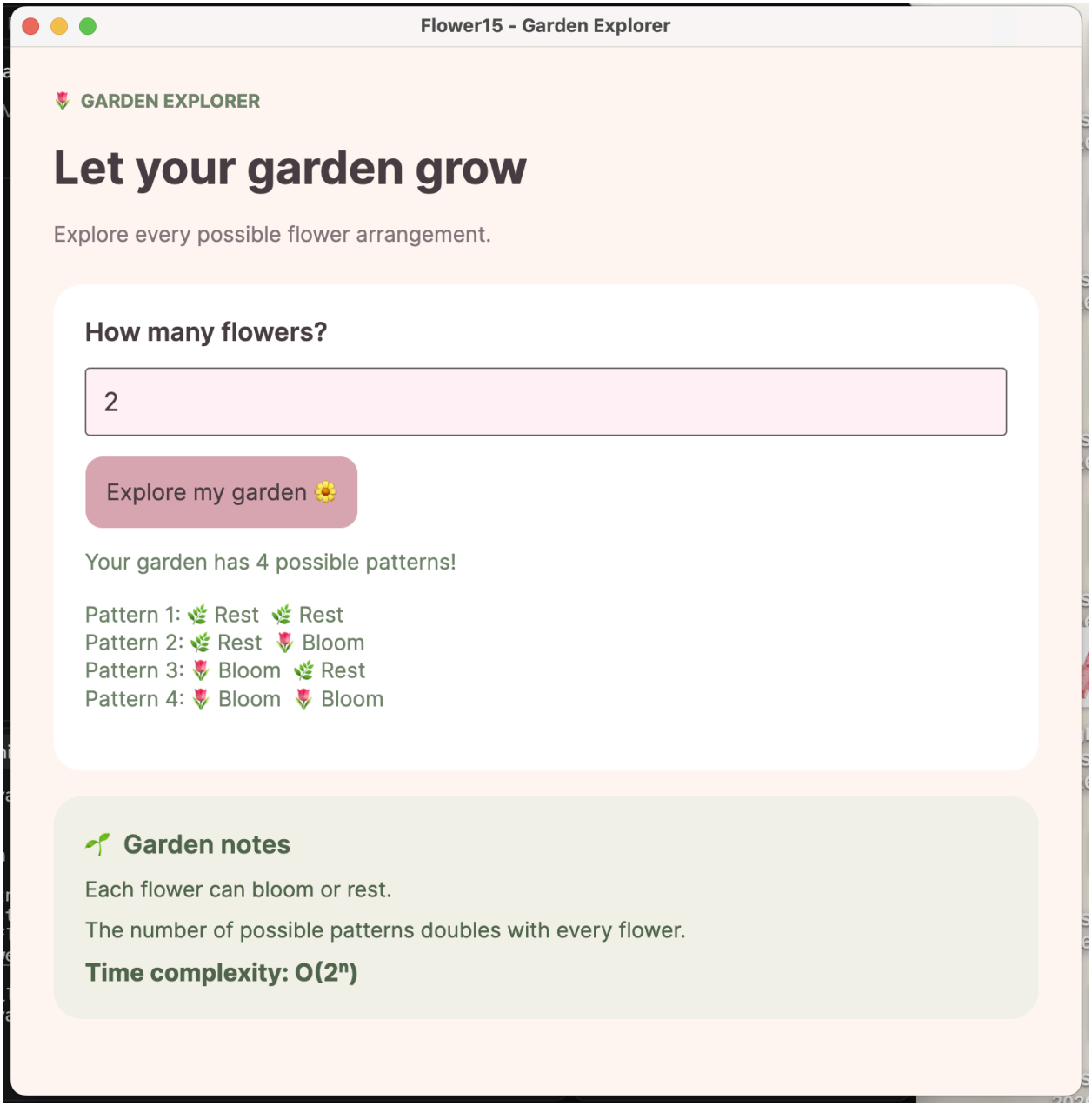

Star11 - Star Pattern Builder
Star Pattern Builder
Create your own little constellation.
NUMBER OF ROWS
8
✦ BUILD STAR PATTERN ✦
YOUR PATTERN
PATTERN DETAILS
Stars created: 36
Made with care and a little stardust.


Star12 - Star Grid Analyzer
✦ STAR GRID ANALYZER ✦
QUADRATIC TIME COMPLEXITY • O(n²)
Enter grid size (1–10)
4
YOUR STAR GRID
★ ★ ★ ★
★ ★ ★ ★
★ ★ ★ ★
★ ★ ★ ★
OPERATION COUNT
16 operations (4 × 4)
✦ GENERATE GRID ✦
NESTED LOOPS • O(n²)

EXPONENTIAL TIME




Petal Combinations
a little garden
Petal Combinations
Choose number of petals (1–6)
3
ALL COMBINATIONS
TOTAL COMBINATIONS
8 combinations
EXPLORE PETALS
RECURSION • O(2ⁿ)


Flower15 - Garden Explorer
GARDEN EXPLORER
Let your garden grow
Explore every possible flower arrangement.
How many flowers?
2
Explore my garden
Your garden has 4 possible patterns!
Pattern 1: Rest Rest
Pattern 2: Rest Bloom
Pattern 3: Bloom Rest
Pattern 4: Bloom Bloom
Garden notes
Each flower can bloom or rest.
The number of possible patterns doubles with every flower.
Time complexity: O(2ⁿ)